

B. Srinivasan

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Profile

Driven AI/ML Engineer and Full-Stack Developer with a strong foundation in designing, training, and deploying scalable machine learning models. Experienced in Natural Language Processing (LLMs, RAG), Computer Vision, and MLOps, with a proven ability to bridge the gap between complex AI architectures and production-ready web applications.

Education

Chennai Institute of Technology

B.E., Computer Science and Engineering (CGPA: 7.98)

Chennai, India

May 2021 – April 2025

– **Relevant Coursework:** Artificial Intelligence, Data Structures & Algorithms, Database Management Systems.

Technical Skills

Data Science & AI: Machine Learning, Deep Learning, NLP, RAG, PyTorch, Hugging Face, OpenCV, YOLO, LangChain

MLOps & Cloud: Docker, MLflow, Git, REST API deployment, Vector Databases (FAISS/Chroma)

Languages: Python, Java, C, C++, SQL, JavaScript, HTML/CSS

Web Development: FastAPI, Flask, Django, React.js, Node.js, Express, Streamlit

Databases: PostgreSQL, MongoDB, MySQL, SQLite

Experience

Web Developer Intern

May 2024 – Jul 2024

S.S.V Traders

Remote

- Developed high-impact website modules using HTML, CSS, and JavaScript with a focus on responsive design across platforms.
- Utilized Material-UI for a modular component library, ensuring a modern and scalable user interface.
- Maintained version control using Git, ensuring seamless integration of features and efficient workflow.
- Mastered backend technologies and optimized server-side logic to support data-heavy application requests.
- Collaborated on API design and integration, ensuring a scalable architecture for cross-functional operations.
- Achieved a **25% reduction in API response times** through efficient database querying and performance tuning.

Projects

Autonomous AI Research Agent | Python, LangChain, LLMs, Prompt Engineering

[GitHub](#)

- Developed an autonomous agent utilizing LangChain to independently scrape, synthesize, and report on web data.
- Implemented asynchronous task execution to handle long-running web searches without blocking the primary user interface.
- Applied advanced prompt engineering and tool-calling to handle complex, multi-step analytical queries.

RAG-Based Document Analyzer | Python, Vector DB, LLMs, Retrieval-Augmented Generation

[GitHub](#)

- Architected a highly accurate RAG pipeline to extract and analyze semantic context from complex PDF documents.
- Implemented efficient document chunking and metadata filtering to enhance retrieval accuracy and minimize LLM hallucination.
- Integrated embedding models and Google APIs to deliver context-aware, generative answers to user queries.

Real-Time Anomaly Detection System | PyTorch, OpenCV, YOLOv8, FastAPI

[GitHub](#)

- Engineered a real-time computer vision pipeline utilizing YOLOv8 for high-speed, high-accuracy object detection.
- Leveraged transfer learning techniques to fine-tune the model on custom datasets, optimizing detection precision.
- Served model predictions via FastAPI, optimizing inference latency for live data streams.

End-to-End MLOps Pipeline | Scikit-Learn, MLflow, Docker, REST APIs

[GitHub](#)

- Architected a complete machine learning lifecycle, systematically tracking model experiments and metrics via MLflow.
- Containerized predictive algorithms using Docker and deployed a scalable inference endpoint for production.

Generative AI Image Synthesizer | Python, PyTorch, Hugging Face Diffusers, Streamlit

[GitHub](#)

- Engineered a Generative AI pipeline utilizing PyTorch and Hugging Face Transformers for text-to-image synthesis.
- Optimized model inference latency and deployed an interactive, user-facing prototype application using Streamlit.

Certifications

Programming & Dev: PCAP – Programming Essentials in Python, The Complete Web Development Bootcamp

Networking (Cisco): CCNA 1: Intro to Networks, CCNA 2: Switching, Routing, CCNA 3: Enterprise Networking

Automation & Security: RPA Citizen Developer Foundation (UiPath Academy), Introduction to Cybersecurity